

Linear Operator Theory

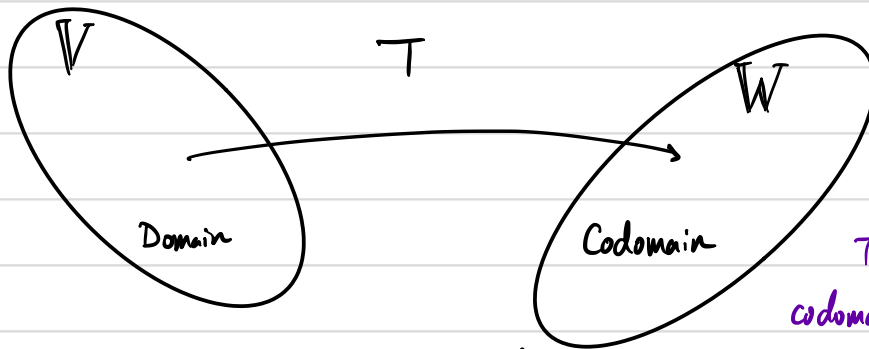
Defn: A linear transformation (or sometimes a linear transform) T on a vector space V to a vector space W is a function $T: V \rightarrow W$ that preserves scalar multiplication and vector addition.

In other words, for all $\vec{u}, \vec{v} \in V$ and $\lambda \in \mathbb{R}$

$$\textcircled{1} T(\lambda \vec{u}) = \lambda T(\vec{u})$$

$$\textcircled{2} T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$$

In this context, the vector space, V , is called the domain for T .
the vector space, W , is called the codomain. Also target.



Pictorial representation of a mapping from one space to another.

There is a difference between codomain and what is tempting to call range. We will discuss range later.

Note: For all linear operators, set $\lambda = 0$, then $T(\vec{0}) = \vec{0}$ ✓

Also, set $\vec{u} = \vec{0}$, then $T(\vec{0} + \vec{v}) = T(\vec{0}) + T(\vec{v})$

$$T(\vec{v}) = T(\vec{0}) + T(\vec{v})$$

$$\Rightarrow T(\vec{0}) = \vec{0} \quad \checkmark$$

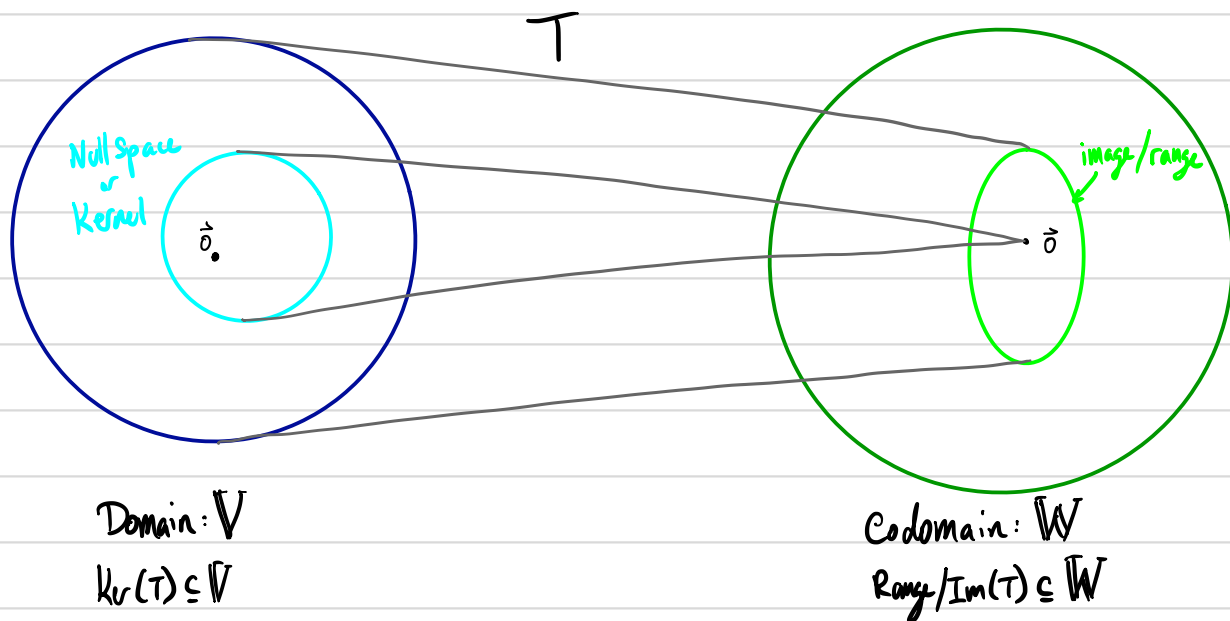
Def'n: The null space, or kernel, of a linear transform $T: V \rightarrow W$ is the set of all vectors in V mapped to the zero vector in W

$$\text{Ker}(T) = \{ \vec{v} \in V \mid T(\vec{v}) = \vec{0} \}$$

Def'n: The image, or range, of a linear transform $T: V \rightarrow W$ is the set of all vectors in W to which T maps the vectors of V .

$$\text{Im}(T) = \{ \vec{w} \in W \mid T(\vec{v}) = \vec{w} \text{ for some } \vec{v} \in V \}$$

Very important! $\text{Ker}(T) \subseteq V$ & $\text{Im}(T) \subseteq W$. It remains to prove that they are subspaces.



Specific Examples of Linear Transforms that come from Matrix Theory

1) Scaling: λI scales a vector by factor λ .

$$\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \lambda x \\ \lambda y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

Note: $\begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} = \lambda^2$ so the area is magnified by λ^2

2) Rotations that fix the origin.

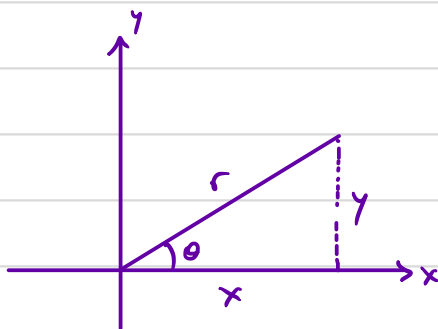
$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}. \quad \text{This matrix rotates a vector by } \frac{\pi}{2} \text{ ccw.}$$

Note: $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (negative identity)

which is strikingly similar to the complex analog

$$i \cdot i = -1$$

In the Complex Plane



$$\text{In } \mathbb{C}, \quad x+iy = r \cos \theta + i r \sin \theta = r (\cos \theta + i \sin \theta) = r e^{i\theta} \quad \checkmark$$
$$x+iy = r e^{i\theta}$$

When mult. any point in $\mathbb{C}$ by $e^{i\phi}$, $e^{i\phi} (r e^{i\theta}) = r e^{i(\theta+\phi)}$,
the effect is rotating that point by angle ϕ . So then

$$e^{i\theta} = 1 \cos \theta + i \sin \theta \leadsto \cos \theta \mathbf{1} + \sin \theta \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \equiv R(\theta)$$

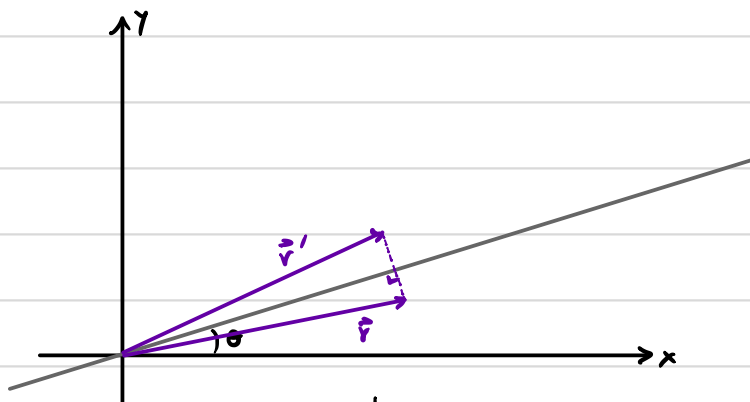
This matrix rotates any vector in $\mathbb{R}^2$ by θ in ccw direction.

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$$

↑
rotation matrix!

Note: $\begin{vmatrix} c & -s \\ s & c \end{vmatrix} = 1$, rotations do not change the area of what they multiply.

3) Reflections about a line forming an angle θ w.r.t. x-axis.



Verify:
$$\begin{pmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$$

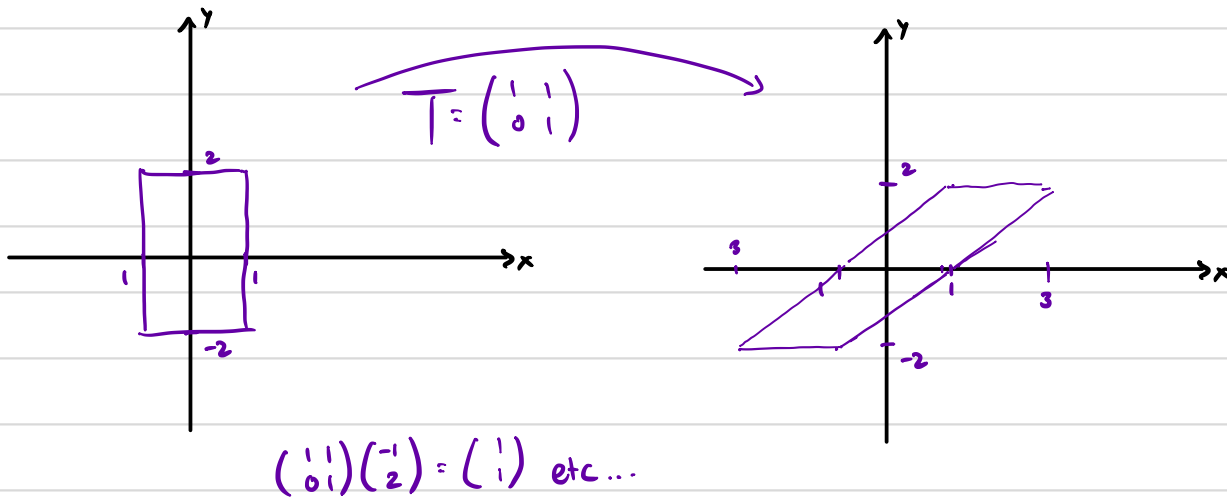
Note: $\begin{pmatrix} c & s \\ s & -c \end{pmatrix} \begin{pmatrix} c & s \\ s & -c \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \leftarrow$ Two reflections leave things unchanged!

Note: Reflecting about $y=x$, $\theta = \frac{\pi}{4}$, so $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix} \quad x \longleftrightarrow y \quad \checkmark$

Note: $\begin{vmatrix} c & s \\ s & -c \end{vmatrix} = -1 \leftarrow$ orientations reversed, no area changes.

4) Shearing: Ex// $\begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + \alpha y \\ y \end{pmatrix}$

Shears by factor α in x -direction the "higher up" one goes.

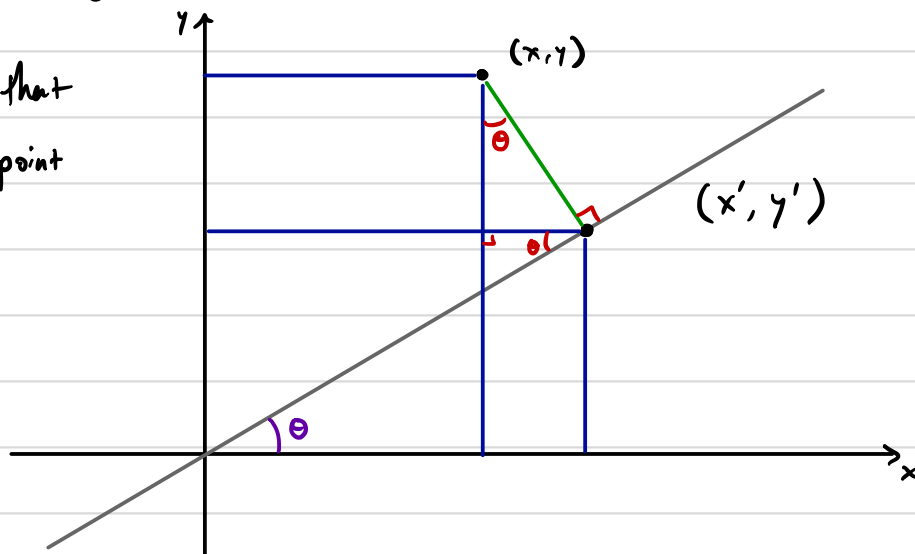


Note: $\begin{vmatrix} 1 & \alpha \\ 0 & 1 \end{vmatrix} = 1$. Shearing does not change areas.

5) Projections onto a line forming an angle θ w.r.t. x -axis.

you're going to prove (w/ help) that
given (x, y) the projected point
will be (x', y') given by

$$\begin{pmatrix} \cos^2 \theta & \sin \theta \cos \theta \\ \sin \theta \cos \theta & \sin^2 \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$$



Note: $\begin{vmatrix} c^2 & sc \\ sc & s^2 \end{vmatrix} = c^2 s^2 - (sc)^2 = 0$, Projections demolish areas. Non-invertible.

It's analogous to being unable to reconstruct a person from their shadow.

6) Dimension changes: An $m \times n$ matrix maps n -dimensional vectors into m -dimensional vectors.

Some other linear transformations

7) Differentiation of a polynomial $\frac{d}{dt}: P_n \rightarrow P_{n-1}$

$$T(\alpha p + \beta q) = \frac{d}{dt}(\alpha p + \beta q) = \alpha p' + \beta q'$$

8) Integration of a polynomial $\int_0^t: P_n \rightarrow P_{n+1}$

$$T(\alpha p + \beta q) = \int_0^t (\alpha p + \beta q) dz = \alpha P(t) + \beta Q(t) - (\alpha P(0) + \beta Q(0)) \quad \checkmark$$

The Standard Matrix for a Linear Transformation

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. The standard matrix associated w/ T is defined by

$$A = [T(\hat{e}_1) \mid T(\hat{e}_2) \mid \cdots \mid T(\hat{e}_n)]$$

Where columns $T(\hat{e}_i)$ are the images of $\hat{e}_i$ under T .

Pf: $T \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = T(v_1 \hat{e}_1 + \cdots + v_n \hat{e}_n) = v_1 T(\hat{e}_1) + \cdots + v_n T(\hat{e}_n)$

$$= \begin{pmatrix} | & & | \\ T(\hat{e}_1) & \cdots & T(\hat{e}_n) \\ | & & | \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = A \vec{v} \quad \checkmark$$

column picture

Ex// Find the matrix of the transformation $T(x, y) = (x - y, x + y, 2x)$

2 input, 3 output! $\therefore m \times n \Rightarrow 3 \times 2$

$$A = [T(\hat{e}_1), T(\hat{e}_2)] = [T\begin{pmatrix} 1 \\ 0 \end{pmatrix}, T\begin{pmatrix} 0 \\ 1 \end{pmatrix}]$$
$$= \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 0 \end{bmatrix} \quad \checkmark$$

check! $\begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 0 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x - y \\ x + y \\ 2x \end{pmatrix}$

Ex// Suppose $T = \frac{d}{dt}$ and we have basis $\{1, t, t^2, t^3\}$

Then if $p(t) = 3 - 2t + 4t^2 + 5t^3$

$$= (3)1 + (-2)t + (4)t^2 + (5)t^3$$

then $p'(t) = (-2)1 + (4)(2)t + (5)(3)t^2 + (6)t^3$

$$A = \begin{bmatrix} | & | & | & | \\ T(1) & T(t) & T(t^2) & T(t^3) \\ | & | & | & | \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{so, } \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ -2 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} -2 \\ 8 \\ 15 \\ 0 \end{pmatrix} \rightarrow -2 + 8t + 15t^2 \quad \checkmark$$

Ex// Can do the same for integration. Is it the inverse of differentiation? The results may surprise you!
[click here for answer!](#) (JK, D-I-Y).

Mapping Theory

Defn: Given a mapping $f: X \rightarrow Y$ and $y \in Y$, the preimage (or inverse image) is the collection of all points in X s.t. $f(x) = y$.

$$\{x \in X \mid f(x) = y\}$$

It is the set of all points that map to y . Also, all the points that y could come from.

Ex// The preimage of $\vec{0}$ is the kernel of f .

Injectivity: A function $f: X \rightarrow Y$ is one-to-one, or injective, provided it is true that $f(u) = f(v) \Rightarrow u = v$.

Note: The direction of implication is important. $u = v \xRightarrow{\text{always}} f(u) = f(v)$, but not always the other way.

Restatement of defn: If $f(u) = f(v)$, then $u = v$

Interpretation: Different inputs give different outputs. If A , then B
If $\sim B$, then $\sim A$.

so if $u \neq v$, then $f(u) \neq f(v)$
for injectivity.

Haik's Redefinition: Every point in the codomain has at most one point in the pre-image.

Ex// $f(x) = x^3$ is injective

Ex// $f(x) = \arcsin(x)$ is injective

Ex// $g(x) = \sin(x)$ is not injective. Note: $\sin(0) = \sin(\pi)$
when $0 \neq \pi$ ✓

Surjectivity: If the image of a function $f: X \rightarrow Y$ is all of Y , the function is said to be onto Y , or to be surjective.

Restatement: $f(X) = Y$

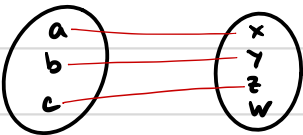
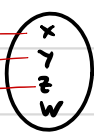
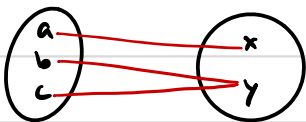
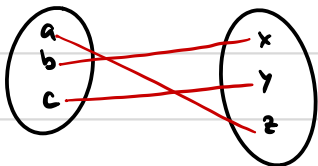
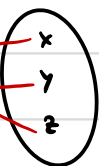
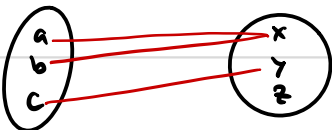
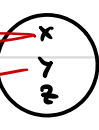
Interpretation: The domain of a function maps to the entire codomain, not just some subset.

Hait's Redefinition: Every point in the codomain has at least one point in the pre-image.

Ex// $f(x) = x$ is surjective

Ex// $g(x) = |x|$ is not surjective. The point -1 in the codomain does not come from a point in the domain.

Pictorial Representations

Domain	Codomain	Injective?	Surjective?
X	Y		
		YES	No
		No	YES
		YES	YES
		No	No

Defn: If a mapping is both injective & surjective, i.e. "one-to-one and onto", it is called bijection.

A function may be an injection, surjection, or a bijection.

(at most) (at least) (exact)

All of this was in general for any kind of function. We will explore this further in the exercises.

A linear mapping that is also a bijection is called an isomorphism.

All isomorphisms are invertible. (Is the backward statement true?)

Isomorphisms are far more important than just in Linear Algebra. They characterize what is meant by "pretty much the same."

Ex// $\mathbb{R} \neq \mathbb{R}^+$ are "isomorphic" through $f(x) = e^x$ & $f^{-1}(x) = \ln(x)$.

Addition in $\mathbb{R}$ is "pretty much the same" as multiplication in $\mathbb{R}^+$

Ex// $T(\vec{v}) = \begin{pmatrix} c & -s \\ s & c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$

The rotation matrix is an isomorphism. The inverse mapping is an equivalent rotation backwards:

$$R^{-1}(\theta) = R(-\theta) = \begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{pmatrix} = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}$$

Note:

$$R R^{-1} = \begin{pmatrix} c & -s \\ s & c \end{pmatrix} \begin{pmatrix} c & s \\ -s & c \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \checkmark$$

check!

Image Theorem

Let $T: V \rightarrow W$ be a linear transform w/ image $\text{Im}(T)$.

(i) $\text{Im}(T)$ is a subspace of W .

(ii) T is surjective if and only if $\text{Im}(T) = W$

pf: (ii) is the definition of surjection.

(i) Suppose $\vec{w}_1, \vec{w}_2 \in \text{Im}(T) \subseteq W$

Then by hypothesis there exists at least one u_1, u_2 s.t.

$$T(\vec{u}_1) = \vec{w}_1$$

$$T(\vec{u}_2) = \vec{w}_2$$

Then

$T(\alpha \vec{u}_1 + \beta \vec{u}_2) = \alpha \vec{w}_1 + \beta \vec{w}_2$, so since $\alpha \vec{u}_1 + \beta \vec{u}_2$ is in the preimage of $(\alpha \vec{w}_1 + \beta \vec{w}_2)$, $\alpha \vec{w}_1 + \beta \vec{w}_2$ is still in W . Thus closure under linear combinations is proven.

Defn: The rank of a linear transformation is given by $\text{rank}(T) = \dim(\text{Im}(T))$.

Ex/ Suppose $T: \mathbb{R}^4 \rightarrow \mathbb{R}^2$

$$T(\vec{v}) = A\vec{v} = \begin{bmatrix} 2 & -4 & 3 & 6 \\ -1 & 2 & -2 & -3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \vec{w}$$

$$\text{Note: } \vec{w} = v_1 \begin{pmatrix} 2 \\ -1 \end{pmatrix} + v_2 \begin{pmatrix} -4 \\ 2 \end{pmatrix} + v_3 \begin{pmatrix} 3 \\ -2 \end{pmatrix} + v_4 \begin{pmatrix} 6 \\ -3 \end{pmatrix}$$

$$\text{The } \text{Im}(T) = \text{span} \left\{ \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \begin{pmatrix} -4 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ -2 \end{pmatrix}, \begin{pmatrix} 6 \\ -3 \end{pmatrix} \right\} \quad !$$

If we perform row reduction,

$$\begin{pmatrix} 2 & -4 & 3 & 6 \\ -1 & 2 & -2 & -3 \end{pmatrix} \quad \frac{1}{2}R_1 + R_2$$

$$\begin{pmatrix} 2 & -4 & 3 & 6 \\ 0 & 0 & \frac{1}{2} & 0 \end{pmatrix} \quad \begin{array}{l} 2R_2 \\ -3R_2 + R_1 \end{array}$$

$$\begin{pmatrix} 2 & -4 & 0 & 6 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad \frac{1}{2}R_1$$

$$\begin{pmatrix} 1 & -2 & 0 & 3 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

↑ ↑

There are two pivot columns that correspond to a basis for the column space of A .

The vectors that correspond to those columns are

$$\left\{ \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ -2 \end{pmatrix} \right\}$$

These form a basis for $\text{Im}(T)$. Thus,

$$\text{rank}(T) = \dim(\text{Im}(T)) = 2 \quad \checkmark$$

T is also a surjection since $\dim(\mathbb{R}^2) = 2$.

// If T is given by a matrix $A \in M_{m \times n}$, then $\text{Im}(T) = \text{Col}(A)$
The pivot columns of A form a basis for $\text{Im}(T)$.

$$\text{rank}(T) = \dim(\text{Im}(T)) = \dim(\text{Col}(A)) = \# \text{ of pivot columns of } A$$

↑

How the I.M.T. joins up again!

The Kernel Theorem

Let $T: V \rightarrow W$ be a linear transform w/ kernel $\text{Ker}(T)$. Then

(i) $\text{Ker}(T)$ is a subspace of V .

(ii) T is injective iff $\text{Ker}(T) = \{\vec{0}\}$

Pf: (i) was done many times, and is also a practice final question.

(ii) If T is injective

$$T(\vec{u}) = T(\vec{v}) \Rightarrow \vec{u} = \vec{v}.$$

so $T(\vec{w}) = \vec{0}$ means $\vec{w} \in \text{Ker}(T)$

$$\text{so } T(\vec{w}) = T(\vec{0}) \Rightarrow \vec{w} = \vec{0}$$

$\therefore$ only $\vec{0} \in \text{Ker}(T)$!

If $\text{Ker}(T) = \{\vec{0}\}$ Suppose $T(\vec{u}) = T(\vec{v})$. Then

$$T(\vec{u}) - T(\vec{v}) = T(\vec{u} - \vec{v}) = \vec{0}$$

This means $\vec{u} - \vec{v} \in \text{Ker}(T)$. But only $\vec{0} \in \text{Ker}(T)$ so $\vec{u} - \vec{v} = \vec{0} \Rightarrow \vec{u} = \vec{v}$ ✓

Ex// Same matrix. $A = \begin{bmatrix} 2 & -4 & 3 & 6 \\ -1 & 2 & -2 & -3 \end{bmatrix}$

The kernel is all $\vec{v}$ st. $A\vec{v} = \vec{0}$. So we want

$$\left[\begin{array}{cccc|c} 2 & -4 & 3 & 6 & 0 \\ -1 & 2 & -2 & -3 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cccc|c} 1 & -2 & 0 & 3 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{array} \right] \quad \begin{array}{l} \text{basic var. : } v_1, v_3 \\ \text{free var. : } v_2, v_4 \end{array}$$

$$\Rightarrow v_3 = 0, \quad v_1 = 2v_2 - 3v_4 \quad \text{so} \quad \vec{v}_h = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} = \begin{pmatrix} 2v_2 - 3v_4 \\ v_2 \\ 0 \\ v_4 \end{pmatrix} = v_2 \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + v_4 \begin{pmatrix} -3 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$